SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Natural Language Processing: Text Embeddings, Transformers & LLMs

Department: CSE AIML	**Course / Subject:** Natural Language Processing (AI701PC)
Semester: Semester 7	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Tokenization, Stemming/Lemmatization, TF-IDF, Word2Vec, GloVe, BERT contextual representations, Prompt Engineering, and Retrieval-Augmented Generation (RAG).

Unit 1: Text Preprocessing & Language Modeling

Tokenization, Stop-word removal, Stemming (Porter), Lemmatization (WordNet), N-gram language models, Perplexity metric, and Part-of-Speech (POS) tagging.

Unit 2: Word Representation Techniques

Bag of Words (BoW), Term Frequency-Inverse Document Frequency (TF-IDF), Word2Vec (Skip-gram & CBOW), Negative Sampling, and GloVe embeddings.

Unit 3: Deep NLP & Sequence Tagging

Bi-directional LSTMs for Named Entity Recognition (NER), Sentiment Analysis with GRU, Sequence-to-Sequence Machine Translation, and Beam Search decoding.

Unit 4: Pre-trained Language Models

Contextual word embeddings (ELMo), BERT masked language modeling, RoBERTa, T5 encoder-decoder, GPT auto-regressive generation, and Fine-tuning paradigms.

Unit 5: Generative AI & Knowledge Graphs

Retrieval-Augmented Generation (RAG) architecture, Vector databases (FAISS, ChromaDB), Prompt Engineering patterns, Hallucination mitigation via Knowledge Graph grounding.